

ECR STAGE-5

APPROVED COMPONENT BASIS

WHOLE-COLUMN ASSEMBLY RECONCILIATION

Design269 · Stage-3 run64 · Stage-4 calculation13 · actual saved Stage-5 record1 / revision1

Component geometry approved and frozen by the user. All approved turbine/stator dimensions are retained. This engineering manifest is not a new production rule-set version. The existing application and historical R1 snapshot remain unchanged.

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION. Retained hydraulic qualification: SCALE_UP_EXTRAPOLATION. The accepted Stage-3 audit was not rerun, reoptimized or reopened.

Disposition	Result
Component freeze	Approved 198-mm shrouded turbine and 112-mm-centre / 84-hole stator frozen as a canonical hashed engineering manifest.
Positioned internals	39 rotors and40 stators reconcile at the actual saved elevations over7020mm centre-plane active span.
Approved-component interferences	None found against positioned vessel, supports and saved connection-neck envelopes using full rotating envelopes. No approved dimension changes.
Assembly gate	Retained interface holds: support-wall conformity, shaft/head penetration, bearings/coupling, plate retention and installation details. These are not silently passed.
Production status	Not implemented; no database or saved drawing changes. Whole-column detail release is not approved by this report.

Component canonical SHA-256: 954fd39266e55dbaa2c0c10dcc6578b5a94d9247b1ff95194f0a762b58a32042

1 · Approval boundary and frozen component manifest

The approval freezes the detailed component geometry from the final engineering closure report. It does not retrospectively modify old R1 rotor/stator dimensions, approve fabrication or qualify the complete assembly. The manifest reproduces the exact approved hole diameter and every hole centre; no display rounding changes its geometry.

Feature	Frozen value [mm unless noted]	Authority
Column ID / rotor OD / pitch / speed	600 / 198 / 180 /45RPM	Inherited Stage3
Turbine architecture	Double-entry radial-flow shrouded;6 straight radial blades at60°	User-approved construction basis
Eye / radial working width	130 /34	Approved preliminary engineering choice; not source correlation
Shaft / hub OD × height	44 /70×24	Approved preliminary component envelope
Blade thickness / inner height / paddle height	3 /16 /28	Approved component choices
Upper and lower shrouds	OD198 / ID130 / thickness2 each	Approved geometry
Overall height	32 =28+2×2	System calculated
Stator OD / plate thickness	600 /4	Approved geometry
Overlap CE / stator centre	9 /112=130–2×9	Approved CE; calculated DSO
Perforations	84 ×Ø39.5594790278	Exact nominal area closure
Rows PCD / quantities	200/12;310/18;420/24;530/30	Approved layout
First angles / increments	15°/30°;10°/20°;7.5°/15°;6°/12°	Approved sixfold stagger
No-hole lanes	Six ×8 wide, axes0/60/120/180/240/300°	Retained plate; no extra blockage
Gross physical free area	0.400000; shaft-only diagnostic0.3946222222	Not returned to upstream correlations

The95% turning-area sizing screen remains removed. Positive overlap is geometric projection, not a sealed duct or proof of hydraulic capture. Actual upper/lower eye-face areas remain11752.698mm² each; hub/web cuts and radial-turn areas are diagnostics, not proven hydraulic throats.

2 · Actual saved assembly and chain verification

Record / verification	Evidence
Saved design / record / revision	269 /1 /1, saved 2026-09-19T18:06:21.098Z
Source Stage3 / Stage4	64 /13; IDs and Stage3 immutable hashes agree
Saved Stage5 snapshot integrity	Canonical snapshot matches database immutable_hash
Saved source integrity	Canonical sourceStage3+sourceStage4 hash matches source_hash
Saved geometry and rules manifest	Both canonical hashes match the immutable snapshot declarations
Stage4 chain	Saved Stage5 sourceStage4 result equals SELECT-read calculation13 result; stored Stage3 hash matches run64
Method	Direct SELECT inside BEGIN READ ONLY, rolled back. No API, optimizer, audit replay or source writer invoked.
Snapshot provenance	Actual database record—not the earlier browser verification fixtures.

The recorded chain is the specifically requested saved basis. Snapshot currentness flags alone are not used to infer hydraulic validity. The preserved accepted upstream audit establishes the choice; this task only verifies record consistency.

Frozen inherited fact	Value
D / DR / hc	600 /198 /180mm
Operating speed / retained window	45RPM /30–60RPM
Phase configuration	NMP continuous / RRBO dispersed
Required / installed active height	7000 /7020mm
Compartment count / separators	39 /40 under existing two-terminal-boundary convention
Qualification	SCALE_UP_EXTRAPOLATION

3 · Coordinate convention and complete stack closure

Z=0 bottom internal head pole; axis X=Y=0; +Z upward; azimuth counterclockwise from +X viewed from +Z; all report lengths mm. The same datum and saved elevations are retained. Turbine-local Z=0 translates to each saved rotor centre; approved shrouds extend ± 16 and stator plates ± 2 .

Assembly datum / interval	Elevation or span [mm]
Skirt bottom / vessel bottom pole	– 300 / 0
Bottom head / bottom tangent	0–150 / 150
Bottom end-zone centre allowance	150–750 = 600
First stator material interval	748–752
First / last rotor centre	840 / 7680
Active separator-centre span	750–7770 = 7020 = 39×180
Last stator material interval	7768–7772
Complete separator-material stack	748–7772 = 7024; not 7020
Top end-zone centre allowance	7770–8370 = 600
Actual terminal-plate-face to head-tangent distance	598 each end
Straight shell tangent span	150–8370 = 8220
Top head / top pole	8370–8520 / 8520
Pedestal / drive body	8520–8670 / 8670–9120
Overall skirt-to-drive height	– 300–9120 = 9420
Shaft extent / length	450–8670 / 8220

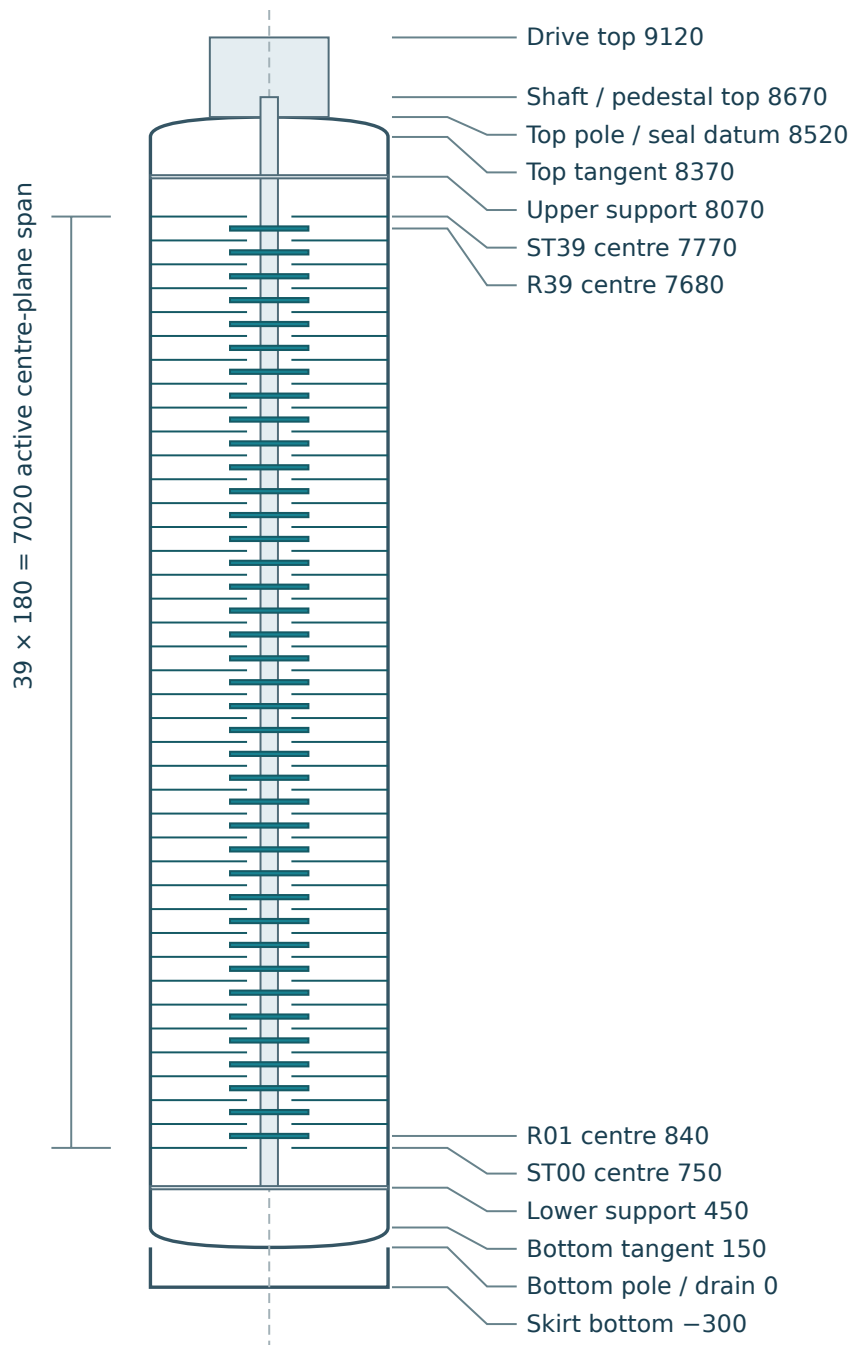
Plate thickness does not add 40×4 to active height: the 40 separator centres already bound 39 pitches. The terminal plates protrude 2mm beyond each active centre-plane boundary. The 39 compartment face clear heights are 176mm; each 32mm rotor leaves 72mm on both sides. Adjacent rotating envelopes have 148mm axial separation.

Old saved rotor height 30.8 becomes approved 32 (+0.6mm per side); old stator thickness 3.75 becomes 4 (+0.125mm per side). Both substitutions are evaluated at unchanged centres. Old single opening 379.473319 is replaced only in this successor study by 112 plus 84 holes; historical data is untouched.

4 · Whole-column review elevation

A-01 · Saved vessel with approved replacement internals

Elevation datum: bottom internal head pole Z = 0; all dimensions mm



Teal: approved successor components; grey: retained saved assembly envelopes.

Axial scale compressed. Plate/rotor visibility enhanced; schedules govern.

5 · All39 turbine elevations and clearances

All lengths mm. R-bottom/top include both shrouds. Adjacent plate face clearances are calculated individually from saved elevations, not inferred from a representative compartment.

Rotor	Centre	Bottom	Top	Lower gap	Upper gap
R01	840	824	856	72	72
R02	1,020	1,004	1,036	72	72
R03	1,200	1,184	1,216	72	72
R04	1,380	1,364	1,396	72	72
R05	1,560	1,544	1,576	72	72
R06	1,740	1,724	1,756	72	72
R07	1,920	1,904	1,936	72	72
R08	2,100	2,084	2,116	72	72
R09	2,280	2,264	2,296	72	72
R10	2,460	2,444	2,476	72	72
R11	2,640	2,624	2,656	72	72
R12	2,820	2,804	2,836	72	72
R13	3,000	2,984	3,016	72	72
R14	3,180	3,164	3,196	72	72
R15	3,360	3,344	3,376	72	72
R16	3,540	3,524	3,556	72	72
R17	3,720	3,704	3,736	72	72
R18	3,900	3,884	3,916	72	72
R19	4,080	4,064	4,096	72	72
R20	4,260	4,244	4,276	72	72
R21	4,440	4,424	4,456	72	72
R22	4,620	4,604	4,636	72	72
R23	4,800	4,784	4,816	72	72
R24	4,980	4,964	4,996	72	72
R25	5,160	5,144	5,176	72	72
R26	5,340	5,324	5,356	72	72

Rotor	Centre	Bottom	Top	Lower gap	Upper gap
R27	5,520	5,504	5,536	72	72
R28	5,700	5,684	5,716	72	72
R29	5,880	5,864	5,896	72	72
R30	6,060	6,044	6,076	72	72
R31	6,240	6,224	6,256	72	72
R32	6,420	6,404	6,436	72	72
R33	6,600	6,584	6,616	72	72
R34	6,780	6,764	6,796	72	72
R35	6,960	6,944	6,976	72	72
R36	7,140	7,124	7,156	72	72
R37	7,320	7,304	7,336	72	72
R38	7,500	7,484	7,516	72	72
R39	7,680	7,664	7,696	72	72

6 · All40 stator elevations and quantities

Stator	Centre [mm]	Bottom face	Top face
ST00	750	748	752
ST01	930	928	932
ST02	1,110	1,108	1,112
ST03	1,290	1,288	1,292
ST04	1,470	1,468	1,472
ST05	1,650	1,648	1,652
ST06	1,830	1,828	1,832
ST07	2,010	2,008	2,012
ST08	2,190	2,188	2,192
ST09	2,370	2,368	2,372
ST10	2,550	2,548	2,552
ST11	2,730	2,728	2,732
ST12	2,910	2,908	2,912
ST13	3,090	3,088	3,092
ST14	3,270	3,268	3,272
ST15	3,450	3,448	3,452
ST16	3,630	3,628	3,632
ST17	3,810	3,808	3,812
ST18	3,990	3,988	3,992
ST19	4,170	4,168	4,172
ST20	4,350	4,348	4,352
ST21	4,530	4,528	4,532
ST22	4,710	4,708	4,712
ST23	4,890	4,888	4,892
ST24	5,070	5,068	5,072
ST25	5,250	5,248	5,252
ST26	5,430	5,428	5,432
ST27	5,610	5,608	5,612

Stator	Centre [mm]	Bottom face	Top face
ST28	5,790	5,788	5,792
ST29	5,970	5,968	5,972
ST30	6,150	6,148	6,152
ST31	6,330	6,328	6,332
ST32	6,510	6,508	6,512
ST33	6,690	6,688	6,692
ST34	6,870	6,868	6,872
ST35	7,050	7,048	7,052
ST36	7,230	7,228	7,232
ST37	7,410	7,408	7,412
ST38	7,590	7,588	7,592
ST39	7,770	7,768	7,772

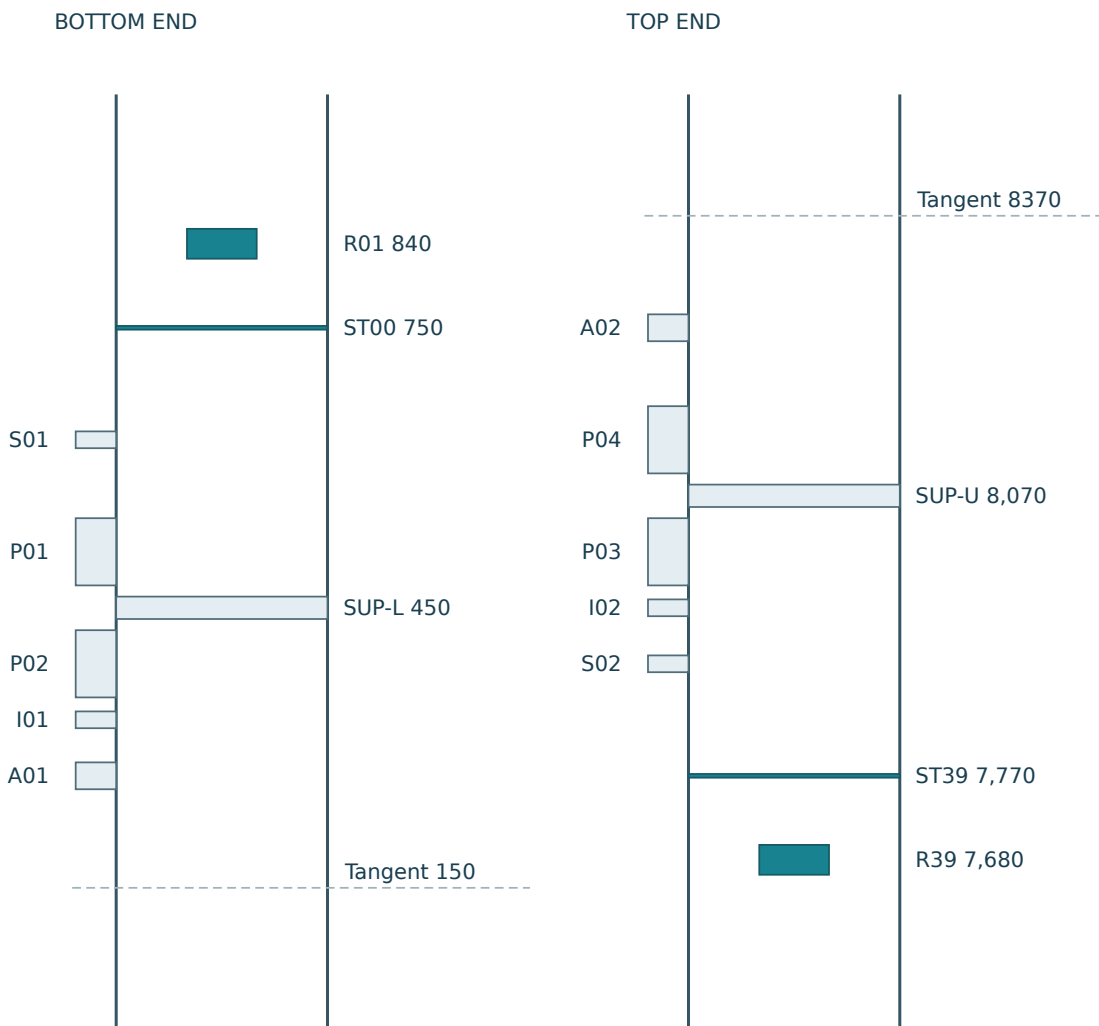
Part	Whole-column quantity
rotors	39
hubs	39
blades	234
shrouds	78
stators	40
statorHoles	3360
shaft	1
shaftSupports	2
supportArms	6
connections	12

Every plate uses the same approved full84hole pattern. Six no-hole lanes are retained parent plate, not additional bars.40plates comprise38 internal shared separators plus2terminal plates; this is the existing assembly convention, not a universal literature requirement.

7 · End zones, shaft supports and rotating clearance

A-02 · End zones and support axial placement

Projected connection OD bands; azimuth and exact centres follow in schedule.



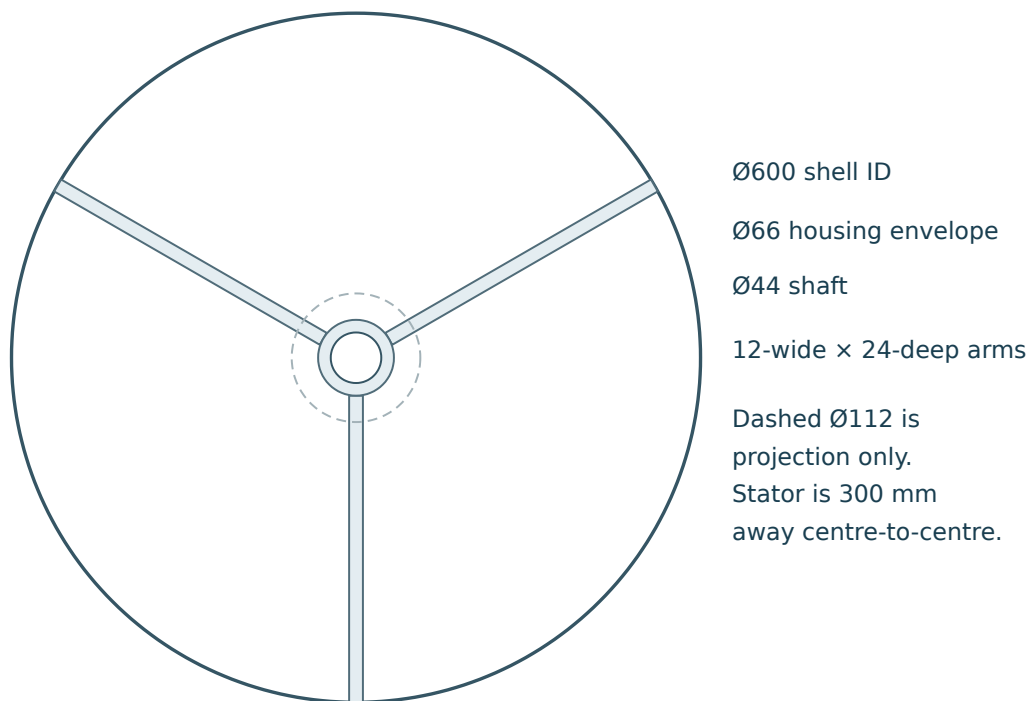
Support bands: 438-462 and 8058-8082. Nearest terminal plate gap: 286.
Support-to-process-neck OD axial margin: 12; nozzle-to-terminal plate: ≥ 109 .
Connections are projected for spacing only; this is not a common cutting plane.

Interface	Minimum geometric clearance [mm]
Support to terminal plate	286
Support to nearest rotating envelope	362
Support to process-connection OD band	12
Nozzle OD band to terminal plate	109
Nozzle OD band to rotor envelope	185
Support to head envelope	288

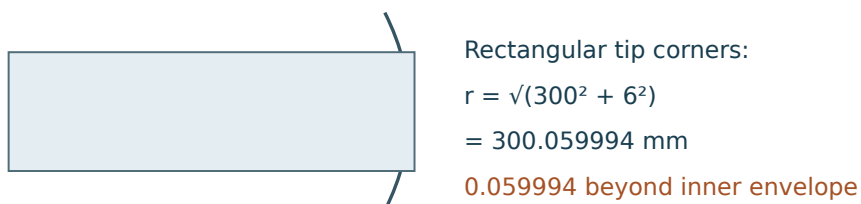
8 · Support plan, wall conformity and shaft engagement

A-03 · Support / shaft / wall interfaces

Saved three-arm supports at Z450 and Z8070; azimuths 30°, 150°, 270°



Wall termination — existing boundary requires detailing



Detail is exaggerated; not to scale. Specify shell-conformal termination / weld.

This is an inherited assembly-interface issue, not turbine/stator interference.

Housing bore, support joints and lower journal engagement remain undefined.

No approved component dimension changed; no historical support solid overwritten.

The approved Ø112 stator opening does not need to pass the Ø66 support housing or radial arms at the same station: their centres are 300mm outside the terminal separator centres, with 286mm face separation. Shrinking the stator central opening therefore causes no support/stator collision.

At each wall end the saved rectangular arm uses $u=300$, $|v| \leq 6$, so its corners lie 0.059994mm outside the specified internal shell radius. This is a real inherited boundary nonconformance, not a new rotor/stator clash. It is not evidence of penetration depth into an actual wall because no wall thickness is specified.

Recommended assembly-only resolution, not silently applied: use a shell-conformal end $u=\sqrt{(300^2-v^2)}$, plus a separately engineered wall attachment. At $|v|=6$ the axial-in-plan trim is 0.060006mm. The hub-side arm root is also a tangent interface rather than a saddle; at $|v|=6$ a conformal $\varnothing 66$ housing surface is $u=\sqrt{(33^2-6^2)}=32.44996$, compared with saved $u=33$. The joint/weld fills or profiles that transition. No approved turbine or stator dimension changes.

Lower shaft starts at 450 inside the 438–462 lower housing band: nominal envelope engagement only 12mm, not 24. Upper housing 8058–8082 lies wholly along the shaft. Neither housing specifies bearing bore, retention or bearing type. These are intended shaft-support interfaces requiring definition, not solid disks certified clear of the shaft.

9 · Full12-connection schedule

All original bores, OD, centres, azimuths and 60mm projections are retained. The ten shell connections project radially outward from r300; no inward dip tube, distributor or quill is part of the saved model. OD bands, not centre lines alone, govern separation.

Tag / service	Axis / azimuth	Centre Z	Bore / OD	Envelope Z range
P01 RRBO feed	radial /0°	510	60 /72	474–546
P02 NMP-rich extract outlet	radial /180°	390	60 /72	354–426
P03 NMP feed	radial /180°	8,010	60 /72	7,974–8,046
P04 RRBO-rich raffinate outlet	radial /0°	8,130	60 /72	8,094–8,166
A01 Low-shell flush	radial /270°	270	24 /28.8	255.6–284.4
A02 High-shell equalization	radial /270°	8,250	24 /28.8	8,235.6–8,264.4
S01 Lower sample connection	radial /90°	630	15 /18	621–639
S02 Upper sample connection	radial /90°	7,890	15 /18	7,881–7,899
I01 Lower instrument connection	radial /45°	330	15 /18	321–339
I02 Upper instrument connection	radial /45°	7,950	15 /18	7,941–7,959
D01 Bottom-head drain	down /axial°	0	24 /28.8	-60–0.173
V01 Top-head crown-region vent	up /315°	8,490	24 /28.8	8,484.246–8,550

D01 is a downward axial drain on the bottom pole. V01 is an upward neck at radial offset 180 and 315°, not an absolute-high-point vent. Both preserve the complete 64-point saved head/neck intersection boundary. Saved bore/OD does not specify pressure-wall thickness, flange, valve or reinforcement geometry.

10 · Connection coordinates and boundary treatment

Tag	Centre XYZ [mm]	Endpoint XYZ [mm]
P01	300, 0, 510	360, 0, 510
P02	-300, 0, 390	-360, 0, 390
P03	-300, 0, 8,010	-360, 0, 8,010
P04	300, 0, 8,130	360, 0, 8,130
A01	-0, -300, 270	-0, -360, 270
A02	-0, -300, 8,250	-0, -360, 8,250
S01	0, 300, 630	0, 360, 630
S02	0, 300, 7,890	0, 360, 7,890
I01	212.132, 212.132, 330	254.558, 254.558, 330
I02	212.132, 212.132, 7,950	254.558, 254.558, 7,950
D01	0, 0, 0	0, 0, -60
V01	127.279, -127.279, 8,490	127.279, -127.279, 8,550

The report calculation JSON retains every stored head-intersection point. All128 points were checked against the analytical ellipsoids with residual below 10^{-8} mm. For D01 the saved outer-neck penetration boundary spans $Z=0.172900$; the downward endpoint is -60 . For V01 it spans 8484.246050 – 8495.076617 , with neck endpoint 8550 . The exact stored head ellipsoid, not a flat plane at neck centre, supplies these boundary limits.

Analytical head convention: bottom $Z=150[1-\sqrt{1-(r/300)^2}]$; top $Z=8370+150\sqrt{1-(r/300)^2}$. Head thickness, material surfaces and cut/weld details are not present. The intended nozzle/head or nozzle/shell penetration itself is not treated as an unintended collision; pressure-boundary construction remains open.

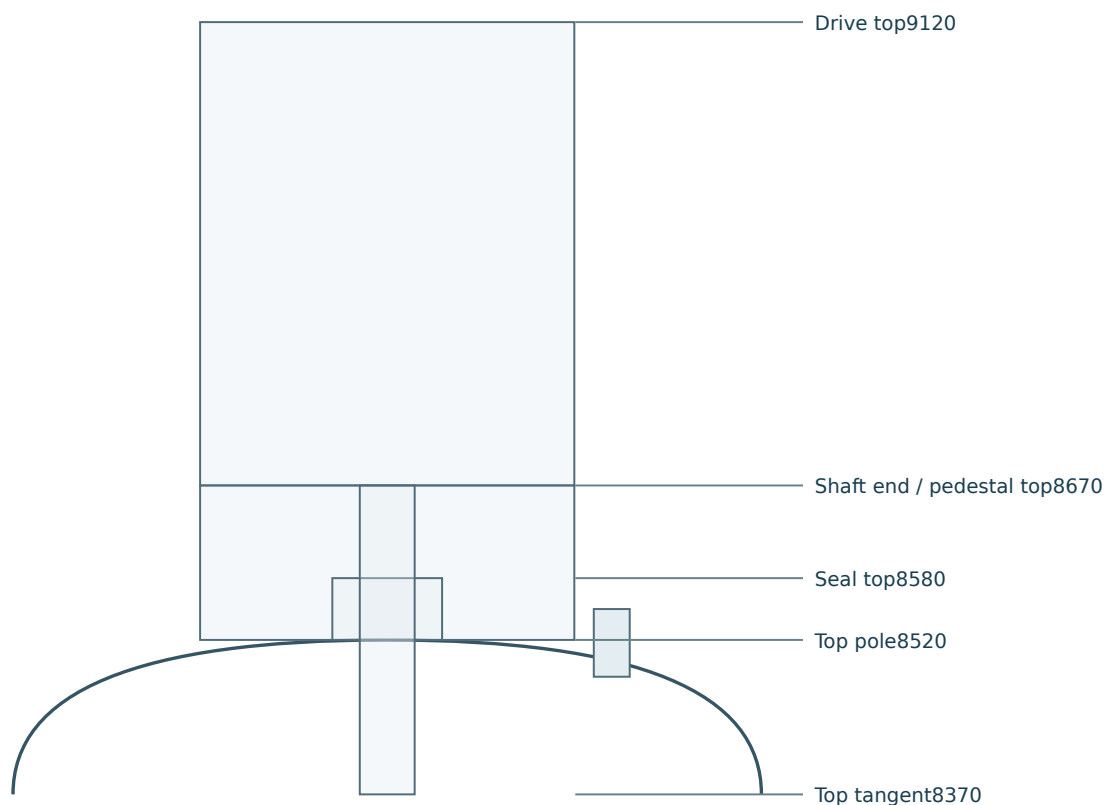
All66 neck pairs have positive clearance between enclosing capsules; minimum lower bound 219.319994 mm. Capsules conservatively enclose the OD neck cylinders and saved axial head-boundary extensions. A positive bound proves separation of these modelled necks only; flanges, valves and field piping have no defined envelopes.

The drain is inside the skirt perimeter, not colliding with a solid $\varnothing 600$ skirt cylinder. Radial drain-to-skirt nominal gap 285.6 . Drain endpoint -60 is 15 mm above the access cutout top -75 ; access window 180 wide $\times$ 150 high is centred at $Z=-150$, azimuth 0 . No tool-removal or maintenance-access claim is made.

11 · Top head, seal and drive integration

A-04 · Top-head, seal and drive reserved interfaces

Schematic axial/radial projection; actual vent is at azimuth 315°.



Vent inner radial edge165.6; pedestal radius150 → 15.6 geometric clearance.

Shaft requires top-head penetration. Seal/pedestal overlap is reserved nesting.

Pressure-boundary cut, mount, cavity and coupling engagement NOT detailed.

The shaft crosses the top-head ideal surface at Z8519.596123 at r22, rising to pole8520 at r0. This is an intended shaft penetration that must be represented as a real cut and sealed boundary in the next assembly detail. The saved snapshot contains no complete solid-resolved shaft neck or pressure-boundary opening.

Seal envelope Ø88 spans8520–8580; pedestal Ø300 spans8520–8670. Their overlap is intentional nested reserved space, not two asserted solid bodies. The shaft ends at8670, exactly where the drive-body envelope begins; coupling insertion/engagement is not defined. The assembly cannot be released as a fully connected solid model on envelope evidence alone.

Vent r180 with OD28.8 has inner radial edge165.6, leaving15.6mm to the Ø300 pedestal/drive cylinder where their Z extents overlap. Its clearance to the Ø88 seal cylinder is121.6. Head mount, shell reinforcement and drive/seal service space remain unqualified.

12 · Exhaustive check matrix and proof boundaries

The standalone calculation writes every row to assembly-reconciliation.json. 4476 checks are enumerated, including intended interfaces. “PASS” means geometric separation for the stated ideal solids/bounds, not mechanical adequacy. Zero intended interfaces are never counted as positive clearance.

Pair family	Count	Minimum [mm]	Status
nozzle bore-OD	12	1.5	PASS
rotor-stator	1560	72	PASS
rotor-rotor	741	148	PASS
stator-stator	780	176	PASS
rotor-support	78	362	PASS
stator-support	80	286	PASS
rotor-nozzle	468	185	PASS
stator-nozzle	480	109	PASS
support-nozzle	24	12	PASS
rotor-shell	39	201	PASS
stator-shaft	40	34	PASS
stator-shell	40	0	INTENDED INTERFACE
rotor-shaft	39	0	INTENDED INTERFACE
nozzle-nozzle	66	219.319994	PASS
vent-envelope	4	15.6	PASS
nozzle-shaft	10	278	PASS
shell-neck-head	10	105.6	PASS
drain-shaft	1	449.8271	PASS
drain-skirt	1	285.6	PASS
drain-access	1	15	PASS
support-head	2	288	PASS

All 1560 rotor/stator pairs, 741 rotor pairs and 780 stator pairs are enumerated. Rotors are checked as full 360° radius 99×height 32 swept cylinders, enclosing shrouds, finite-thickness blades and hubs. Frozen blade clocking alone is not used as a clash test.

Axial separation of complete bounding intervals is a sufficient proof even when radial projections overlap. The reported axial gaps are separating-plane distances, not necessarily shortest three-dimensional surface distances. No sample meshes or sparse angle sampling are used to claim clearance.

Counterparts represented only by reserved envelopes remain explicitly marked as interfaces. The matrix is not a blanket pass for missing flange, bearing, weld, fastener, fillet, wall or attachment solids.

13 · Assembly attachment and installation reconciliation

Load path / interface	Reconciled location	Remaining assembly definition
Turbine torque path	Six webs connect hub to outer paddles; shrouds meet paddle faces at ± 14	Shaft/hub key or clamp; blade/shroud/hub joint solids and tolerances. No arbitrary hardware inserted.
Stator load path	Plate to continuous peripheral unperforated rim; nominal OD600 at shell ID600	Ledge/retainer or welded seat, bypass seal, fit allowance and access sequence.
Peripheral stator rim	Outer hole edge maximum r284.779740; unperforated rim to r300 is 15.220260	An available geometric region, not approved shelf width or strength calculation. Any added feature must avoid holes.
Radial no-hole lanes	Six 8 wide lanes retained in plate	Not separate bars or supports crossing the central opening. No extra blockage deduction.
Support load path	$\varnothing 66$ housing → three 12×24 arms → shell at Z450/8070	Shell-conformal arm ends, housing bore, journal engagement and wall attachment.
Shaft/seal/drive path	Shaft 450–8670; seal at 8520; pedestal to 8670	Top-head cut/seal mounting, coupling engagement and retained bearing access.
Installation/removal	39 rotors, 40 full-width plates, one continuous shaft	No demonstrated insertion route through an unspecified open flange/head. Assembly sequencing or split/welded construction requires an explicit decision.

Whole-column physical layout reconciliation is complete for the defined ideal envelopes; detailed interface closure is not. The approved components can be retained unchanged. Do not invent a bore, flange, split plate or shaft extension and call it already approved.

Potential additions must be checked against the preserved manifest. Prefer changing assembly interfaces rather than approved turbine/stator geometry. A future request to alter an approved dimension must identify a measured/modelled interference and its location.

14 · Findings, holds and recommended disposition

Finding	Disposition
Approved turbine/stator component basis	FROZEN engineering manifest; no dimensions changed.
39×180 active stack /7020 centre span	Reconciled against actual Stage4 and saved Stage5 elevations.
Rotating and stationary component separation	No unintended interference in the represented positioned volumes.
Inherited support-wall corner overrun	0.059994mm beyond internal shell; assembly-only conformal termination/joint detail required.
Shaft/head and support-housing passages	Intended interfaces, not fully modelled cuts/bearing solids; remain explicit assembly holds.
Stator edge and attachment	Zero nominal fit; fit/retention/bypass seal and installation method unresolved.
Fabrication qualifications	Pressure design, shaft dynamics/critical speed, stresses, fatigue, weld design, balance, corrosion, material selection, tolerances and component selection remain outside this geometric proof.
Hydraulic performance	SCALE_UP_EXTRAPOLATION retained; unchanged areas do not demonstrate successor hydraulic equivalence or pumping capacity.

Recommendation: retain the approved turbine/stator manifest without dimensional changes. Accept the reconciled positioning/envelope layout as the assembly-development basis, subject to closing the named inherited interfaces. Do not declare the entire column fabrication-ready or implement a production successor version from this report alone.

This report distinguishes assembly geometry holds from mechanical/fabrication qualification. The wall-tip boundary, required head penetration and unspecified installation/retention interfaces are actual geometry/detail matters, not issues that may be dismissed merely by a NOT FOR FABRICATION warning.

15 · Traceability, preservation and reproducibility

Identity	SHA-256
sourceHash	0e7f1590065042ca1cf0e6dd1008fe74d505a9658dfa2fab1068aee49a4f490d
immutableHash	33319e763593b53ad942eef313ce73cc98cb0a88c49eea12d8ac1ecd26f6842a
geometryHash	4550be70163150bc87c7dfa78083438f719f63731d64afe1a4f9c0ed3bae0864
stage3ImmutableHash	e2afc059444f3c22c5d3a64b83ed81b59b5a122bd51ac1c9bc9897337d788249
stage4LineageHash	5a9a6446a4e59faff9d5a86ee02f21afba11b1524e02641d5dd449424d380b3
evidenceCanonicalHash	4391b4972bf089dcf47ac5768729e061399ffe263125205bf5441c08bdf2ed43

Approved component manifest canonical SHA-256:

954fd39266e55dbaa2c0c10dcc6578b5a94d9247b1ff95194f0a762b58a32042. Canonicalization recursively sorts object keys, retains array order and hashes UTF-8 JSON.stringify output. The .sha256 file deliberately labels this canonical method; it is not the pretty-printed-file byte hash.

Artifacts: approved-component-manifest.json and .sha256; saved-evidence.json / saved-evidence-after.json; assembly-reconciliation.json with allchecks/schedules/connection boundaries; read-evidence.ts (SELECT-only); reconcile.mjs (offline); build-report.mjs (presentation only); preservation-verification.json and inspection.json.

Reproduce offline: node deliverables/ecr-assembly-reconciliation/reconcile.mjs, then node deliverables/ecr-assembly-reconciliation/build-report.mjs. The database reader is separate and never called by these scripts. No prior report/drawing is overwritten.

The before/after evidence comparison and 68 scientific/presentation source-file hashes are recorded separately. The accepted Stage3 audit and its replay were not rerun. No production successor version, application change or database mutation was performed.

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION. Component approval is distinct from whole-column detailed-assembly and fabrication approval.